Energy-Efficient 200 Gbit/s Transceivers for Optical Access Networks

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Chapter 1

Access Networks and Coherent PON

1.1 Optical Access Networks

The access network is the final link between an operator's central exchange and the end user, often called the *last mile*. Exponentially growing bandwidth demands from streaming, cloud services, and 5G backhaul are placing severe pressure on this segment.

A passive optical network (PON) uses a point-to-multipoint (P2MP) topology in which a single optical line terminal (OLT) at the exchange serves multiple optical network units (ONUs) at the customer premises through a tree of passive optical splitters, as illustrated in fig. 1.1. No active components are required in the distribution plant, keeping operational costs low.

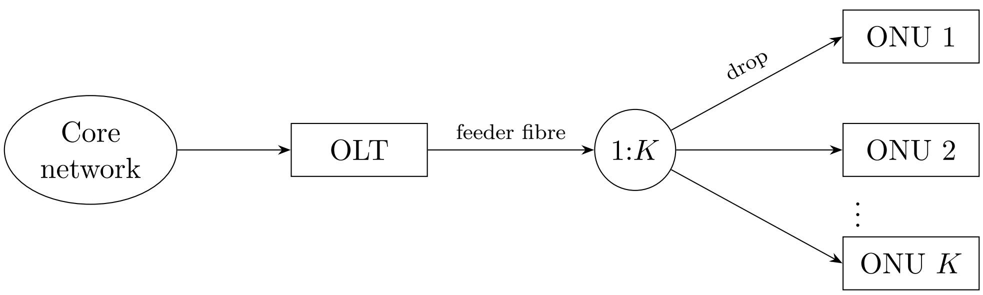


Figure 1.1: Passive optical network: one OLT serves K ONUs via a passive 1:K splitter.

1.2 Coherent PON

Traditional intensity-modulation direct-detection (IM-DD) passive optical networks are approaching their practical limits at data rates approaching 50 Gbps per wavelength. Coherent technology encodes information in the amplitude and phase of the optical signal, allowing modulation formats like QPSK that can carry more bits per symbol. Coherent receivers are also more sensitive, allowing for lower transmit powers. These factors make them a promising alternative to achieve rates of 200 Gbps on an acceptable power budget [7].

1.3 Specification

The Coherent Passive Optical Network (CPON) specification [1] defines a 100 Gbps single-wavelength P2MP system. This will be used as an example to test receiver implementations. This report focuses on the *downstream* direction (OLT to ONU) where receiver optimisations can be shared by all ONUs; the key physical-layer parameters are described in the following sections.

1.3.1 Modulation and Symbol Rate

The downstream signal uses non-differential DP-QPSK at a symbol rate of $R_s = 30.5\,\text{GBd}$. Each polarisation carries 2 bits per symbol by mapping consecutive bit pairs to in-phase and quadrature amplitudes $I, Q \in \{-1, +1\}$, giving QPSK constellation points at $\pm 1 \pm j$. Training and pilot symbols use the same mapping but at amplitude ± 3, placing them outside the data constellation for unambiguous identification. Soft-decision forward error correction (FEC) is used with a 22% coding overhead.

1.3.2 Downstream Frame Structure

The fundamental framing unit is the *DSP subframe* of 3 712 symbols, organised as 116×32-symbol blocks. Every block begins with a pilot symbol; the first block also carries an 11-symbol training sequence, whose first symbol doubles as pilot index 1.

1.3.3 Key System Tolerances

The standard supports a 35 dB loss budget from OLT to ONU. To account for impairments in the core network, an SNR of 35 dB at the OLT must be supported. The OLT transmitter laser must be within $\pm 1.5\,\text{GHz}$ of its DWDM channel centre, and the ONU must acquire signals within $\pm 1.8\,\text{GHz}$ of the grid. In the worst case, the combined transmitter and LO offsets yield a carrier frequency offset (CFO) of up to $\approx 3\,\text{GHz}$ that the receiver DSP must correct. Both OLT and ONU lasers must have a linewidth of at most 1 MHz. The sampling frequency offset (SFO) between OLT and ONU can be at most 40 ppm. The ONU must tolerate a chromatic dispersion of up to 1600 ps/nm and a polarisation mode dispersion of up to 8 ps to support reaches up to 80 km.

The specification requires a pre-FEC bit error ratio of $\leq 2 \times 10^{-2}$, which the soft-decision LDPC FEC reduces to a post-FEC BER of $\leq 10^{-12}$.

Chapter 2

Network and Receiver Models

2.1 Simulation Framework

All DSP algorithms in this report are designed, characterised, and compared through software simulation in MATLAB. The simulated receiver processes DP-QPSK symbols generated at the CPON downstream parameters described in chapter 1.

2.1.1 Fixed-Point Arithmetic

The receiver is evaluated with fixed-point arithmetic, as used in ASIC designs. This captures the effect of quantisation error on each implementation's performance, and underpins a bit-accurate energy model that goes beyond algorithmic complexity. All fixed-point modelling uses the MATLAB Fixed-Point Designer toolkit.

2.1.2 System Parameters

The network under investigation assumes an end-to-end OLT-to-ONU loss budget of 35 dB and a back-to-back transmitter SNR of 35 dB, both taken from the CPON downstream specification of chapter 1. The central wavelength is assumed to be 1550 nm. The chromatic dispersion, polarisation-mode dispersion, and pulse-shaping models used throughout the receiver chain are introduced in the chapters that use them.

2.2 Energy Estimation

The energy-efficiency of a receiver implementation cannot be determined from the receiver energy alone. A simple architecure may reduce receiver energy but incur an SNR penalty for acceptable operation due to a drop in accuracy. This calls for a system-level model of the energy consumption where an SNR penalty can be converted to an equivalent increase in transmitter energy, allowing receiver implementations to be compared by their combined energy.

2.2.1 Receiver Energy

The energy consumption of the receiver can be estimated from the number of real multiplications N_M and real additions N_A it performs to process each symbol. The true cost of these operations depends on the hardware implementation. The reference values shown in table 2.1 provide a useful estimate to compare DSP algorithms but a more accurate estimate would require power simulations for an ASIC implementation.

Table 2.1: Energy cost of arithmetic operations for different bit widths in 45 nm CMOS [9].

n (bits)	Add (pJ)	Multiply (pJ)
8	0.03	0.2
32	0.1	3.1

Dependence on Word Length A ripple-carry adder performs n bit-wise additions on words of length n, so its energy scales linearly with n. A shift-and-add multiplier accumulates n partial products of length n and so scales as n^2. Modern adders and multipliers (e.g. carry-lookahead, Booth/Wallace) reduce these costs by a constant factor or to $n \log n$, so the scalings used here are upper bounds; comparisons between algorithms remain valid because all are penalised equally. The proportionality constants are calibrated by a least-squares fit through the origin to the two data points of table 2.1, giving

$$E_A(n) = 3.16\, n \text{ fJ}, \qquad E_M(n) = 3.03\, n^2 \text{ fJ}. \tag{2.1}$$

Energy consumption for arithmetic operations in modern process nodes is not widely reported, and Dennard scaling (the rule that transistor power density remained constant as they were scaled down) [5] can no longer be applied to extrapolate from the values in table 2.1. As an approximation, the $\approx 75\%$ reduction in total power dissipation at constant frequency reported for an Intel core moving from 45 nm to 14 nm CMOS [17] is taken as a proxy for the per-operation energy scaling. This conflates contributions from leakage, clocks, caches and I/O with the arithmetic energy of interest, so the resulting numbers should be read as an estimate rather than a calibrated figure. Applying this reduction to the 45 nm fit gives

$$E_A(n) = 0.79\, n \text{ fJ}, \qquad E_M(n) = 0.76\, n^2 \text{ fJ}. \tag{2.2}$$

Amortisation Some DSP algorithms perform computations that are reused over multiple symbols. For example, carrier frequency offset (CFO) varies much slower than the symbol rate so can be estimated less frequently. For a computation that requires N_M multiplications and N_A additions and is used by S symbols, the operations per symbol would be N_M/S and N_A/S.

CORDIC Coherent receivers perform phase estimates and rotations on complex-valued data in several DSP blocks. This is done efficiently in fixed-point hardware with the CORDIC algorithm, which evaluates these operations using only shifts and additions. CORDIC iteratively rotates a vector by the fixed angles $\arctan(2^{-i})$, each implemented as a bit-shift and an add/subtract. Each iteration adds roughly one bit of angular resolution, so n bits of precision require n CORDIC iterations and the cost is taken to be equivalent to an n-bit real multiplication.

Total Energy and Power Many DSP blocks operate on complex-valued data; each complex multiplication is counted as 4 real multiplications and 2 real additions, and each complex addition as 2 real additions. The energy per symbol for the entire receiver is then

$$E_{\mathrm{rx},s} = \sum_{k=1}^{L} 2\left[N_{A,k}E_A(n_k) + N_{M,k}E_M(n_k)\right], \tag{2.3}$$

where the receiver is decomposed into L blocks that use different word lengths n_k according to their required precision, and $N_{A,k}$, $N_{M,k}$ are the addition and multiplication counts per symbol. The factor of 2 is for both polarisations. Blocks operating on the oversampled signal at rate $f > R_s$ have their per-sample operation counts scaled by the oversampling ratio f/R_s. The receiver power is then

$$P_{\mathrm{rx}} = R_s E_{\mathrm{rx},s}, \tag{2.4}$$

and converting to energy per bit allows fair comparison across modulation schemes and symbol rates,

$$E_{\mathrm{rx}} = \frac{E_{\mathrm{rx},s}}{\log_2(M)}, \tag{2.5}$$

where M is the modulation order.

2.2.2 Transmitter Energy

The downstream link considered here consists of a single OLT transmitting through a 1:K passive splitter to K ONUs, as shown in fig. 1.1. The OLT power is divided equally between branches, so each ONU receives a $1/K$ share.

Minimum Required SNR The CPON standard specifies a maximum pre-FEC bit-error rate (BER) of 10^{-2} that must not be exceeded for the forward error-correction (FEC) scheme to function correctly. Each receiver implementation therefore has a minimum required SNR to meet this FEC threshold. This SNR, together with the noise characteristics of the optical network, determines the minimum transmitter power.

Noise The relationship between SNR and transmitter power P_{tx} depends on the dominant source of noise in the network. Optical access networks typically operate over short lengths at low power, so amplifiers can be omitted and non-linear noise neglected, leaving shot noise as the dominant impairment. The shot noise power referred to the optical input of a balanced coherent receiver is $2h\nu B_{\mathrm{eff}}$, where the factor of 2 captures the contributions from the signal and LO arms of the balanced detector summed in quadrature, and B_{eff} is the receiver electrical bandwidth, set equal to the symbol rate by a matched bandpass filter after the 1:K splitter in fig. 1.1. The power per polarisation received by each ONU is $P_{\mathrm{tx}}/2\Gamma K$, where Γ is the loss from OLT to ONU. This gives a per-ONU NSR of

$$\mathrm{NSR} = \frac{4\Gamma K h\nu B_{\mathrm{eff}}}{P_{\mathrm{tx}}} + \mathrm{NSR}_0, \tag{2.6}$$

where $\mathrm{NSR} = 1/\mathrm{SNR}$ and NSR_0 is the back-to-back NSR at the OLT. Rearranging gives the transmitted power as

$$P_{\mathrm{tx}} = \frac{4\Gamma K h\nu B_{\mathrm{eff}}}{\mathrm{NSR} - \mathrm{NSR}_0}. \tag{2.7}$$

Assuming Nyquist pulse-shaping and matched bandpass filtering at the receiver ($B_{\text{eff}} = R_s$), the energy per bit transmitted is

$$E_{\text{tx}} = \frac{P_{\text{tx}}}{2KR_s \log_2(M)} = \frac{2\Gamma h\nu}{(\text{NSR} - \text{NSR}_0)\log_2(M)}. \tag{2.8}$$

2.2.3 Combined Energy

Before calculating a combined energy contribution, the transmitted optical energy in eq. (2.8) must be converted into an energy drawn from the electrical power supply by estimating the efficiencies of the transmitting laser and the modulator. The dual-polarisation modulator (fig. 2.1) consists of two nested-Mach-Zehnder IQ modulators sharing a common laser through a polarisation beam splitter, with their outputs recombined by a polarisation beam combiner. Each IQ modulator has the transfer function [16]

$$\frac{E_{\text{output}}(t)}{E_{\text{input}}(t)} = \frac{1}{2}\sin\left[\frac{v_I(t)\pi}{2V_\pi}\right] + \frac{j}{2}\sin\left[\frac{v_Q(t)\pi}{2V_\pi}\right], \tag{2.9}$$

where each MZM is biased at its transmission null so that $v_I = v_Q = 0$ produces no output and the drive voltages swing symmetrically about 0 V.

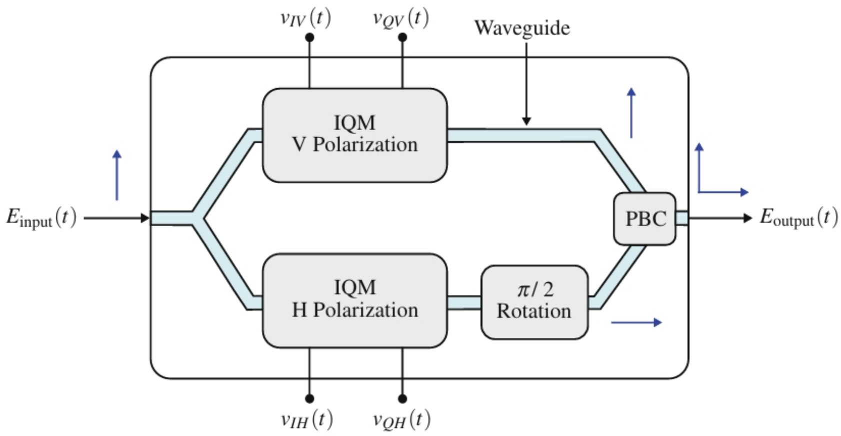


Figure 2.1: Dual-polarisation coherent modulator using MZMs.

QPSK is generated by driving each MZM between adjacent transmission peaks, $v_I(t), v_Q(t) \in \{-V_\pi, +V_\pi\}$, so

$$\frac{E_{\text{output}}(t)}{E_{\text{input}}(t)} = \pm\frac{1}{2} \pm \frac{j}{2} \tag{2.10}$$

and $P_{\text{out}} = P_{\text{in}}/2$, where the loss comes from power coupled to unused ports in the 2×2 couplers. An ideal polarisation beam splitter and combiner introduce no further loss, so the *marginal* efficiency of the dual-polarisation modulator is

$$\eta_{\text{DPM}} = \frac{dP_{\text{tx}}}{dP_{\text{in}}} = 0.5. \tag{2.11}$$

The overall efficiency would also include the power dissipated in the drive electronics, but this is independent of transmit power and so cancels in a comparison of receiver implementations. Combining with the wall-plug efficiency of the laser η_{WP} gives an overall efficiency of

$$\eta_{\mathrm{T}} = \eta_{\mathrm{WP}}\eta_{\mathrm{DPM}} = 0.5\eta_{\mathrm{WP}}. \tag{2.12}$$

Recently reported values for the wall-plug efficiency of lasers suited to coherent systems are 16-19% [19, 18] which gives an overall efficiency of just under 10%. A measure of combined energy is then

$$E_{\mathrm{sys}} = \frac{1}{\eta_{\mathrm{T}}}E_{\mathrm{tx}} + E_{\mathrm{rx}}. \tag{2.13}$$

This does not give the total energy consumed by the optical network, but a difference in E_{sys} shows the energy savings across receiver implementations.

Chapter 3

Equalisation and Clock Recovery

3.1 Introduction

The receiver must compensate two distinct channel impairments and recover the sampling phase before symbol decisions are made. Chromatic dispersion (CD) is static and accurately characterised by the fibre parameters, so it is removed by a fixed filter derived analytically from the channel model. Polarisation-mode dispersion (PMD) varies on millisecond timescales and is unknown a priori, so it is tracked by an adaptive equaliser that also absorbs residual inter-symbol interference (ISI) from the transmit pulse shape. Independently, the receiver samples the incoming waveform with an unsynchronised clock that drifts relative to the transmitter, leaving a fractional timing error that must be estimated and corrected.

The static, adaptive, and clock-recovery blocks share the same data stream and can be arranged in several ways. A Gardner timing error detector (TED) operating in the time domain naturally pairs with a time-domain CD filter or directly with the adaptive equaliser; a Modified Godard TED operates in the frequency domain and can re-use the same FFT as the overlap-save CD equaliser, eliminating the FFT cost of the timing recovery loop entirely. This chapter derives both equaliser stages and both timing recovery algorithms, then combines them into two end-to-end implementations and compares their BER and energy performance: Gardner DPLL followed by the adaptive equaliser, and overlap-save CD/matched filtering with an embedded Modified Godard loop followed by the adaptive equaliser.

Filter lengths and tap counts are sized for the worst-case CPON tolerances of section 1.3.3: 1600 ps/nm of CD, 8 ps of PMD, and a 30.5 GBd symbol rate at a central wavelength of 1550 nm.

3.2 Equalisation

3.2.1 Channel Impairments

Chromatic dispersion CD arises because the group velocity of light in fibre depends on wavelength, so the frequency components of a pulse accumulate a frequency-dependent delay that spreads them over many symbol periods. It is an all-pass effect with a quadratic phase response, simulated by transforming the signal to the frequency domain, applying the transfer function

$$H_{\mathrm{CD}}(\omega) = \exp\left(j\,\frac{D\lambda^2}{4\pi c}\,L\,\omega^2\right), \tag{3.1}$$

where D is the dispersion coefficient, λ the central wavelength, L the fibre length, and c the speed of light, then transforming back with an inverse FFT.

Polarisation-mode dispersion PMD is a frequency-dependent group delay between the two principal states of the fibre caused by random birefringence along the link, simulated as a concatenation of N_{pmd} birefringent sections, each applying a differential group delay of $\pm\tau/2$ to its two principal states in the frequency domain, with a random unitary rotation (the singular vectors of a complex Gaussian matrix) modelling the mode coupling between sections. The mean DGD scales as $\tau \propto D_{\mathrm{PMD}}\sqrt{L}$, with D_{PMD} the fibre PMD coefficient. PMD must be distinguished from *polarisation mixing*, the essentially frequency-flat unitary rotation of the polarisation pair that arises from any uncompensated rotation between the transmit and local-oscillator reference frames; the adaptive equaliser must track both, but only PMD requires multiple taps.

RRC pulse shaping The transmit pulse is a root-raised-cosine (RRC) with roll-off β and a finite truncation span; the matched filter at the receiver is identical which removes ISI and rejects noise. A larger β makes the time-domain pulse shorter and easier to truncate, allowing for fewer taps in the matched filter which reduces complexity. A span of 10 symbols with $\beta = 0.2$ is reasonable for an access network where bandwidth requirements are less strict and low complexity is critical.

3.2.2 Static Equalisation

Chromatic dispersion is compensated by applying the inverse of eq. (3.1),

$$H_{\mathrm{eq}}(\omega) = \exp\left(-j\,\frac{D\lambda^2}{4\pi c}\,L\,\omega^2\right), \tag{3.2}$$

whose inverse Fourier transform is a finite-energy chirp

$$h(t) = \sqrt{\frac{c}{jD\lambda^2 L}}\;\exp\left(j\,\frac{\pi c\,t^2}{D\lambda^2 L}\right). \tag{3.3}$$

The equaliser may be realised either as a time-domain FIR per polarisation, with taps obtained by sampling eq. (3.3) at the receiver period T and truncating to N_{CD} coefficients, or in the frequency domain. As the data stream is continuous, the frequency-domain filter is applied block-by-block by overlap-save: each block of N samples is transformed with an FFT, multiplied by the precomputed samples of H_{eq}, and inverse transformed. Discarding the first $N_{\mathrm{CD}} - 1$ samples of every block removes the circular-convolution wrap-around, leaving $N - N_{\mathrm{CD}} + 1$ valid output samples per block.

3.2.3 Adaptive Equalisation

PMD is tracked with a 2×2 MIMO equaliser comprising four FIR filters whose outputs are

$$x_{\mathrm{out}} = \mathbf{h}_{xx}^H \mathbf{x}_{\mathrm{in}} + \mathbf{h}_{xy}^H \mathbf{y}_{\mathrm{in}}, \quad y_{\mathrm{out}} = \mathbf{h}_{yx}^H \mathbf{x}_{\mathrm{in}} + \mathbf{h}_{yy}^H \mathbf{y}_{\mathrm{in}}. \tag{3.4}$$

The error terms are produced blindly using the constant modulus algorithm (CMA), which exploits the fact that an ideal QPSK constellation lies on a circle of fixed radius. CMA minimises $J = \mathbb{E}\Big[(|x_{\text{out}}|^2 - R)^2\Big]$, where the dispersion constant $R = \mathbb{E}[|s|^4]/\mathbb{E}[|s|^2]$ reduces to a constant for QPSK. The stochastic gradient yields

$$e_x = R - |x_{\text{out}}|^2, \quad e_y = R - |y_{\text{out}}|^2, \tag{3.5}$$

and the filter weights are updated as

$$\begin{aligned} \mathbf{h}_{xx} &\leftarrow \mathbf{h}_{xx} + \mu e_x \mathbf{x}_{\text{in}} x^*_{\text{out}}, \quad & \mathbf{h}_{xy} &\leftarrow \mathbf{h}_{xy} + \mu e_x \mathbf{y}_{\text{in}} x^*_{\text{out}}, \\ \mathbf{h}_{yx} &\leftarrow \mathbf{h}_{yx} + \mu e_y \mathbf{x}_{\text{in}} y^*_{\text{out}}, \quad & \mathbf{h}_{yy} &\leftarrow \mathbf{h}_{yy} + \mu e_y \mathbf{y}_{\text{in}} y^*_{\text{out}}, \end{aligned} \tag{3.6}$$

with step size μ restricted to powers of two for hardware simplicity. CMA suffers from the singularity problem, in which both outputs converge to the same input; this is avoided by initialising $\mathbf{h}_{xx}$ with a single one at the centre tap and seeding the y-polarisation weights from the conjugate-symmetric relation $\mathbf{h}_{yx}[n] = \mathbf{h}^*_{xx}[-n]$, $\mathbf{h}_{yy}[n] = -\mathbf{h}^*_{xy}[-n]$ once x_{out} has converged [14]. CMA also has the advantage that it is robust to phase rotations caused by CFO or phase noise which simplifies the DSP architecure by avoiding feedback from the carrier recovery block.

Parallelism. At a symbol rate of 30.5 GBd the equaliser must produce more than one output per CMOS clock period, so it is parallelised across P lanes. Each lane $p \in \{0, \ldots, P-1\}$ owns its own input buffer

$$\mathbf{x}_p = [\mathbf{x}[mP+p],\ \mathbf{x}[mP+p-1],\ \ldots,\ \mathbf{x}[mP+p-(N-1)]] \tag{3.7}$$

and forms its own output and error using eqs. (3.4) and (3.5), but a single shared set of filter weights is held constant across the P lanes of a block. Each lane contributes its own gradient term and the weights are then updated once per block from the summed contribution; for $\mathbf{h}_{xx}$ this is

$$\mathbf{h}_{xx} \leftarrow \mathbf{h}_{xx} + \mu \sum_{p=0}^{P-1} e_x[p]\, \mathbf{x}_p\, x^*_{\text{out}}[p], \tag{3.8}$$

and analogously for the three other filters.

Sign-Sign CMA. A variant of the CMA update replaces the error and output terms by their signs: $e \to \text{sgn}(e)$ and $x_{\text{out}} \to \text{csgn}(x_{\text{out}})$, where the complex signum function is defined as

$$\text{csgn}(z) = \text{sgn}(\Re z) + j\ \text{sgn}(\Im z). \tag{3.9}$$

This removes all multiplications at no performance cost [2] so will be employed in the later analysis.

3.2.4 Matched Filtering

The receiver maximises signal-to-noise ratio at the symbol decision instants by filtering the received signal with a matched filter, the time-reversed conjugate of the transmit pulse. For the RRC pulse used here the matched filter is itself an RRC; the cascade of transmit and receive RRCs is a raised cosine that satisfies the Nyquist criterion. Two implementations are possible. The matched filter may be implemented explicitly as a frequency-domain mask multiplied into the overlap-save CD stage at no additional cost: $H_{\mathrm{eq}}(\omega)$ is replaced by $H_{\mathrm{eq}}(\omega)H_{\mathrm{RRC}}(\omega)$ and the same overlap-save block performs CD compensation and matched filtering simultaneously. Alternatively the matched filter can be folded into the taps of the adaptive equaliser, provided the adaptive FIR is long enough to span both the matched-filter response and the channel memory it tracks; this saves the explicit filter but requires a longer adaptive FIR.

3.2.5 Filter Length

The CD equaliser must span the channel memory introduced by dispersion. An experimentally verified length for several pulse shapes is given by [10]

$$N_{\mathrm{CD}} = \left\lceil \frac{6.67|\beta_2|LR_s}{T} \right\rceil, \tag{3.10}$$

where T is the receiver sampling period. With an oversampling rate of 2 and the worst-case CPON tolerance of 1600 ps/nm at 1550 nm ($|\beta_2| \approx 20.4\,\mathrm{ps}^2/\mathrm{km}$), this yields a 22-tap filter. The associated overlap-save FFT size is chosen as the power of two that minimises the per-symbol multiplication count, $N = 128$.

The adaptive equaliser must span the channel memory introduced by PMD. The mean differential group delay is $\langle \Delta\tau \rangle = D_{\mathrm{DGD}}\sqrt{L}$, and the instantaneous DGD follows a Maxwellian distribution [15]

$$\mathbb{P}[\Delta\tau > x] \approx \frac{4}{\pi}\frac{x}{\langle \Delta\tau \rangle}\exp\left(-\frac{4x^2}{\pi\langle \Delta\tau \rangle^2}\right). \tag{3.11}$$

For an outage probability of 10^{-12} and the CPON tolerance of 8 ps total PMD, the required span is well under one $T = 16.4$ ps sampling period, so it can be ignored and a single-tap adaptive equaliser can be employed to perform polarisation demultiplexing if matched filtering and CD compensation have already been performed.

3.2.6 Computational Complexity

A radix-2 FFT of size N decomposes into $\log_2 N$ stages of $N/2$ butterflies, each costing 4 real multiplications and 6 real additions. Each overlap-save block needs a forward FFT, an inverse FFT of equal cost, and N complex multiplications by the precomputed filter spectrum, shared over $N - N_{\mathrm{CD}} + 1$ valid outputs. The FFT can be simplified further by rounding each twiddle factor to the nearest signed power of two in its real and imaginary parts, replacing every twiddle multiplication with a pair of bit-shifts and eliminating the real multiplications inside the FFT and IFFT entirely. The per-symbol counts of the time-domain FIR, overlap-save, and overlap-save with power-of-two twiddles are summarised in table 3.1, where $\eta = f_s/R_s$ is the receiver oversampling factor.

Implementation	RM / symbol	RA / symbol
Time-domain FIR	$4\eta N_{\mathrm{CD}}$	$\eta(4N_{\mathrm{CD}} - 2)$
Overlap-save (FD)	$\eta \dfrac{4N \log_2 N + 4N}{N - N_{\mathrm{CD}} + 1}$	$\eta \dfrac{6N \log_2 N + 2N}{N - N_{\mathrm{CD}} + 1}$
Overlap-save with 2^k twiddles	$\eta \dfrac{4N}{N - N_{\mathrm{CD}} + 1}$	$\eta \dfrac{6N \log_2 N + 2N}{N - N_{\mathrm{CD}} + 1}$

Table 3.1: Per-symbol real multiplication (RM) and real addition (RA) counts for the three CD equaliser implementations. N is the overlap-save FFT size, N_{CD} the filter length from eq. (3.10), and η the receiver oversampling factor.

The per-symbol cost of the adaptive equaliser is dominated by the four FIR outputs and the weight update, which both scale with the filter length N. The direct update of eq. (3.6) costs four complex multiplications per tap; the sign-sign reformulation of section 3.2.3 replaces these by sign manipulations and is multiplier-free. Per-symbol counts for both variants are given in table 3.2; the parallelisation of section 3.2.3 amortises the single update over P symbols and so does not appear in the per-symbol count.

Stage	RM / symbol	RA / symbol
FIR outputs (4 FIRs $\times$ N taps)	$8N$	$4N + 2$
Error terms	2	1
Weight update – direct	$8N + 2$	$8N$
Weight update – sign-sign	0	$8N$

Table 3.2: Per-symbol real-operation counts for the parallel-lane butterfly CMA equaliser with N taps per FIR. Sign-sign refers to the multiplier-free update described in section 3.2.3.

3.3 Clock Recovery

3.3.1 Gardner Timing Recovery

The Gardner TED computes an error term from samples $x[n]$ at $\eta = 2$ as [20]

$$e[k] = \Re\{x^*[2k-1]\,(x[2k] - x[2k-2])\}\,. \tag{3.12}$$

The sample $x[2k-1]$ sits midway between two symbol-rate samples; when timing is locked, this midpoint vanishes and $e[k] = 0$, while a small timing error makes the midpoint take the sign of the slope between $x[2k-2]$ and $x[2k]$. The error feeds a proportional-integral (PI) DPLL whose output drives a cubic Farrow interpolator [6]that re-samples the input. The loop uses the same P-lane parallelisation as the adaptive equaliser (section 3.2.3): each lane runs its own interpolator with a per-lane fractional interval, but all lanes within a block share the same PI output so the TED contributions of every lane can be summed before the loop filter is updated.

3.3.2 Modified Godard Timing Recovery

An RRC pulse has bandwidth $(1+\beta)/T$, so for the Fourier transform of the received signal $X(f)$ the product $C(f) = X(f)X^*(f-1/T)$ is non-zero only where the spectra overlap, and is real-valued in this region due to the symmetry of the RRC spectrum. A timing offset τ introduces a phase ramp in the frequency domain, so the overlap becomes

$$C(f) = X(f)X^*\big(f - \tfrac{1}{T}\big)\ e^{-j2\pi\tau/T}, \tag{3.13}$$

whose phase is independent of f and encodes the timing offset. A discrete equivalent for oversampling factor η is given by [12]

$$S = \sum_{k=k_{\text{lo}}}^{k_{\text{hi}}} X[k]\, X^*[k + (1-1/\eta)N], \quad k_{\text{lo}} = \tfrac{1-\beta}{2\eta}N, \quad k_{\text{hi}} = \tfrac{1+\beta}{2\eta}N, \tag{3.14}$$

where the summation is restricted to the bins covered by the RRC excess band. The timing error normalised to the sampling period is then

$$\hat{\tau}_s = -\frac{\eta}{2\pi}\arg(S), \tag{3.15}$$

which is fed into a PI loop filter; the next block applies the accumulated τ_s as a frequency-domain phase ramp on the FFT output before the IFFT. Because S is formed from the same FFT used by the overlap-save CD/matched-filter stage, the Modified Godard TED adds no additional FFT cost.

3.3.3 Computational Complexity

The two detectors operate on blocks of different sizes – Gardner on a block of P output samples at $\eta = 2$ (i.e. $P/2$ symbols), Modified Godard on an overlap-save block of N FFT bins producing $N - N_{\text{CD}} + 1$ valid samples (i.e. $(N - N_{\text{CD}} + 1)/\eta$ symbols) – so the per-block counts are normalised to per-symbol counts in table 3.3, counting complex multiplications as 4 RM + 2 RA and complex additions as 2 RA. The Gardner cost is dominated by one cubic Farrow interpolator per output sample, evaluated as four fixed-coefficient 4-tap FIRs on complex data (one per polynomial coefficient $c_0, \ldots, c_3$) followed by evaluation of the polynomial in μ, giving 38 RM and 30 RA per output sample. The Godard cost is dominated by the $\beta N/\eta$ complex products of eq. (3.14) and the per-block frequency-domain phase ramp (N recursive ramp coefficients and N spectral-bin multiplications, $\approx 8N$ RM); both are amortised over the $(N - N_{\text{CD}} + 1)/\eta$ valid symbols delivered by the overlap-save block. The FFT and IFFT used by Godard are reused from the overlap-save CD/matched-filter stage and so appear in table 3.1 rather than here.

Stage	RM / symbol	RA / symbol
Gardner TED	4	8
Gardner Farrow interpolator	76	60
Godard metric S (eq. (3.14))	$\frac{4\beta N}{N - N_{\text{CD}} + 1}$	$\frac{4\beta N}{N - N_{\text{CD}} + 1}$
Godard arg(S) (CORDIC)	$\frac{\eta}{N - N_{\text{CD}} + 1}$	0
Godard FD phase ramp	$\frac{8N\eta}{N - N_{\text{CD}} + 1}$	$\frac{4N\eta}{N - N_{\text{CD}} + 1}$

Table 3.3: Per-symbol real-operation counts for the two clock-recovery detectors. Gardner runs at $\eta = 2$ and shares the parallelisation of section 3.2.3; the per-lane Farrow and TED costs collapse to a fixed per-symbol count after dividing by $P/2$ symbols per block. Modified Godard runs once per overlap-save block of N samples, amortising over $N - N_{\text{CD}} + 1$ valid output samples (the overlap-save overhead). The shared FFT/IFFT used by Godard is accounted for in table 3.1.

3.4 Combined Implementations

A common architecture for short range access-network receivers is to use a single butterfly adaptive equaliser to perform all of the equalisation in one 2×2 MIMO structure [13]. However, for a longer network like the one under investigation, the filter must be at least $N_{\text{CD}} = 22$ taps long (eq. (3.10)), or up to $22 + 2 \times 10 - 1 = 41$ taps for a conservative equaliser that fully compensates CD and the RRC span.

At those lengths a time-domain FIR is no longer the cheapest implementation, and both the static and the adaptive equalisers become candidates for a frequency-domain implementation. Replacing the static CD/matched-filter FIR with an overlap-save FFT block drops the per-symbol multiplication count from $O(N)$ to $O(\log_2 N)$ (table 3.1), and the resulting FFTs are then available for the Modified Godard TED. A frequency-domain adaptive equaliser [8] achieves the same per-symbol FFT scaling but the MIMO structure is wasteful when PMD is negligible, and the Modified Godard TED cannot be placed after it as the output is at the symbol rate. This motivates two combined implementations: An overlap-save static equaliser to perform CD compensation and matched filtering followed by either the Godard (in the frequency domain) or Gardner (after the IFFT) and finally a single-tap adaptive equaliser for polarisation demultiplexing.

3.5 Results & Discussion

3.5.1 Combined Performance

Table 3.4 compares the performance of the proposed implementations with floating point precision in the absence of CFO. The conservative static equaliser length of 41 seems to offer little improvement over 22 taps, hence 22 taps can be used to reduce complexity. Rounding the FFT/IFFT twiddle factors to a power of 2 incurs an SNR penalty but should still be considered as the receiver's SNR floor may be determined by other DSP blocks.

In reality a CFO can be present in the signal as the carrier recovery block is downstream of equalisation and clock recovery. The effect of CFO on performance is shown in fig. 3.1. CFO is clearly problematic for both clock recovery algorithms, with none able to achieve the 3 GHz tolerance, but the Godard TED struggles the most. The issue can be seen from eq. (3.14); the summation used to perform the estimate is over specific frequency bins but a CFO shifts the spectrum and the sum no longer captures the intended overlap. The effect of a CFO Δf on Gardner is more subtle but can be seen by expressing the signal with CFO as $y[k] = x[k] \exp(j2\pi\Delta f T\eta k)$ then substituting into eq. (3.12) to give

$$e[k] = \Re\left\{x^*[2k-1]x[2k]\exp(j2\pi\Delta f T\eta) - x^*[2k-1]x[2k-2]\exp(-j2\pi\Delta f T\eta)\right\}. \tag{3.16}$$

Taking expectations and noting that the correlation of QPSK symbols is real, we obtain

$$\mathbb{E}[e[k]] = \cos(2\pi\Delta f T\eta)\left\{\mathbb{E}[x^*[2k-1]x[2k]] - \mathbb{E}[x^*[2k-1]x[2k-2]]\right\}, \tag{3.17}$$

so a CFO flattens $\mathbb{E}[e[k]]$ (known as the S-curve), making it harder for the loop to track errors.

Block	N_{SEQ}	N_{AEQ}	2^k twiddles	SNR @ FEC Limit (dB)
Gardner	22	1	no	3.98 ± 0.24
Gardner	22	1	yes	8.47 ± 0.30
Gardner	41	1	no	3.93 ± 0.23
Gardner	41	1	yes	8.41 ± 0.34
Godard	22	1	no	4.41 ± 0.30
Godard	22	1	yes	7.04 ± 0.15
Godard	41	1	no	4.32 ± 0.30
Godard	41	1	yes	6.40 ± 0.12

Table 3.4: SNR @ FEC limit (mean ± standard deviation over trials) for each combined equalisation/clock recovery configuration in the absence of CFO. $N_{\text{SEQ}}, N_{\text{AEQ}}$ are the number of taps in the static and adaptive equalisers respectively.

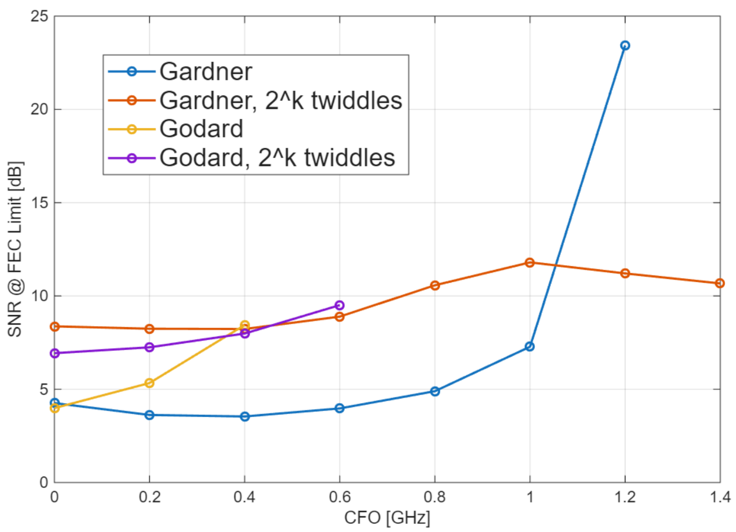


Figure 3.1: SNR @ FEC limit as CFO varies for each implementation, plotted points stop when the FEC limit is no longer reached at any SNR.

Correcting CFO An energy-efficient coarse CFO correction can be built from the FFT already produced by the overlap-save CD/matched-filter. The CFO estimate is found from the centroid of the FFT magnitudes, combined over multiple blocks to improve the estimation accuracy.

$$\hat{k} = \frac{\sum_k k\,|X[k]|^2}{\sum_k |X[k]|^2}, \qquad k \in \{0, \ldots, \tfrac{N}{2} - 1, -\tfrac{N}{2}, \ldots, -1\}, \tag{3.18}$$

with bin spacing $\eta R_s/N$, and applied as a continuous-valued time-domain phasor $\exp(-j2\pi\hat{k}n/N)$ to later samples before they reach the FFT. This estimator is applied before the matched filter as it is band-limited and biases the estimator closer to DC. Figure 3.2 shows the SNR at the FEC limit for each implementation. The coarse estimate allows both Godard and Gardner to achieve the FEC limit at the 3 GHz tolerance but the residual CFO is still impactful for Godard. 2^k twiddles consistently underperform the standard FFT/IFFT for Gardner but towards the 3 GHz limit using them with Godard appears to offer the same performance within the given variance but saves energy.

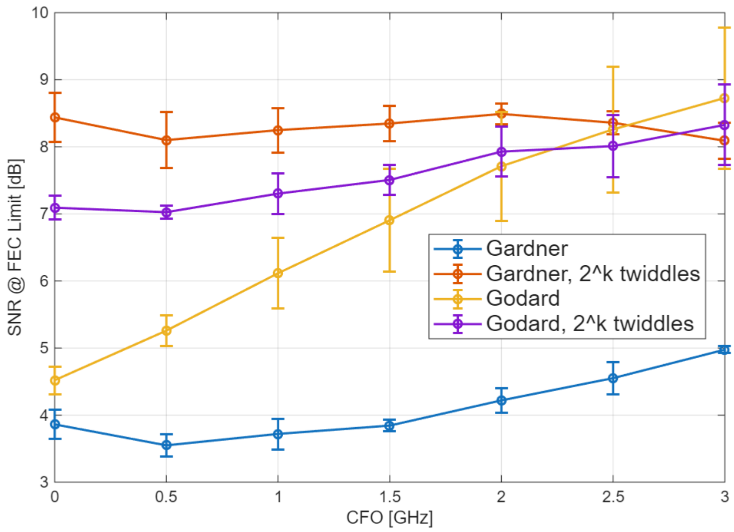


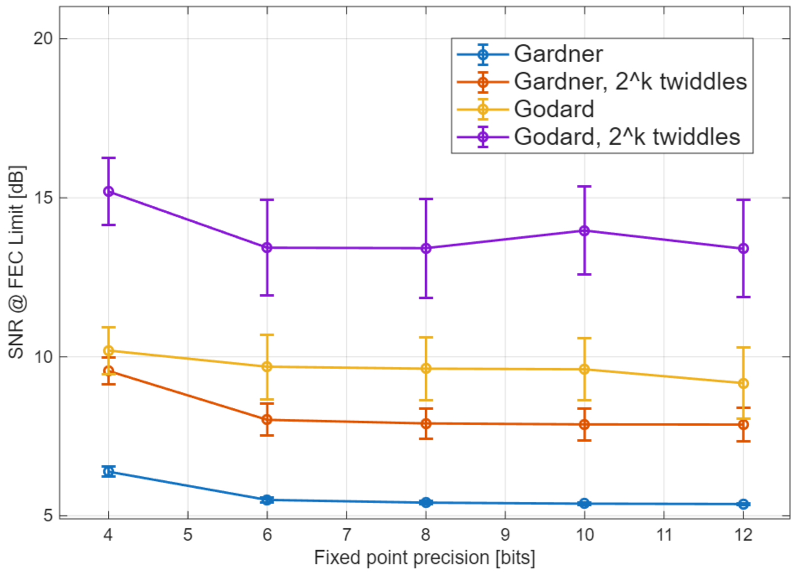
Figure 3.2: SNR @ FEC limit as CFO varies with coarse CFO correction.

3.5.2 Fixed Point Precision

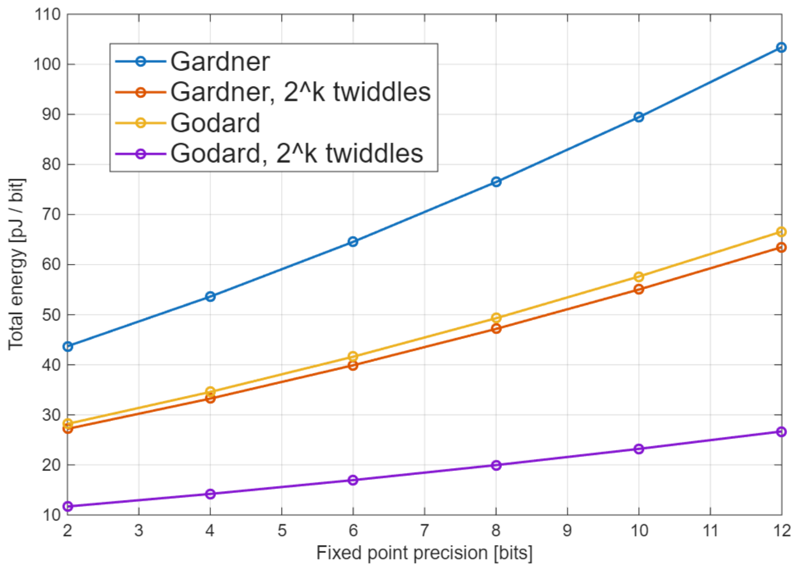
So far implementations have been compared in floating point arithmetic, which avoids the quantisation error present in a real coherent receiver. Figure 3.3a shows the performance for a bit-accurate model of each combination. The most notable change from the SNRs at 3 GHz in fig. 3.2 is for Godard with 2^k twiddles, increasing by ≈ 5 dB and no longer making it a viable choice. Gardner with a standard FFT/IFFT still offers the best performance and from fig. 3.3b, Gardner with 2^k twiddles narrowly beats Godard as the low energy alternative. Precision can go as low as 6 bits for both Gardner implementations with a small rise in SNR at 4 bits, thus the combinations most promising for an energy-efficient receiver are shown in table 3.5. Clock recovery and static equalisation dominate the energy consumption but using 2^k significantly reduces the contribution from the static equaliser.

2^k twiddles	Precision (bits)	SNR @ FEC (dB)	E_{stat}	E_{aeq}	E_{clk}	E_{tot}
no	4	6.39 ± 0.16	24.94	3.28	25.39	53.61
yes	4	9.55 ± 0.42	4.57	3.28	25.39	33.24
no	6	5.50 ± 0.08	29.99	3.94	30.61	64.54
yes	6	8.02 ± 0.50	5.35	3.94	30.61	39.90

Table 3.5: Energy breakdown (pJ/bit) and SNR at the FEC limit for the Gardner implementation with and without 2^k twiddles at 4 and 6 bits of fractional precision (the same precision is used in every block). E_{stat}, E_{aeq}, and E_{clk} are the per-bit energies of the static FD CD/matched-filter, the adaptive equaliser, and the clock-recovery loop respectively.



(a) SNR @ FEC limit vs fixed-point precision.



(b) Energy per bit vs fixed-point precision.

Figure 3.3: Performance and energy of each implementation as fixed-point precision varies with 3 GHz of CFO applied.

Oversampling rate. It is important to note the impact of performing this analysis with an oversampling rate of 2. Modern coherent receivers operate below 2 samples per symbol to reduce the load on analogue-digital converters (ADCs). A standard adaptive equaliser requires 2 samples per symbol as it effectively splits the signal into odd and even samples, which is not possible with fractional oversampling rates. This means the output of the ADC must eventually be upsampled but the clock recovery can be done at the lower sampling rate if Modified Godard is used, reducing the operation counts per symbol in table 3.3. Thus for lower oversampling rates Godard may be the preferred choice.

Chapter 4

Carrier Recovery

4.1 Introduction

A coherent receiver recovers the transmitted symbols by mixing the incoming optical signal with a local oscillator (LO) laser, allowing the amplitude and phase of each symbol to be extracted directly. However, the transmitter and LO lasers will in general have different centre frequencies and independent phase trajectories, producing a carrier frequency offset (CFO) and accumulated phase noise on the received symbols. Carrier recovery is the DSP stage responsible for estimating and removing these impairments before symbol decisions are made.

Carrier recovery is divided into two sequential stages. Frequency recovery performs a finer estimate of the CFO than discussed in chapter 3 and corrects it so that only slow residual phase variations remain. Phase recovery then tracks and removes these residual phase errors on a block-by-block basis.

Two common carrier recovery algorithms, Rife & Boorstyn and Viterbi & Viterbi, are compared against low-complexity alternatives to quantify the trade-offs between energy consumption and performance in an energy-efficient carrier recovery block.

4.2 Frequency Recovery

4.2.1 Carrier Frequency Offset

A carrier frequency offset (CFO) occurs because the transmitter and local-oscillator lasers are free-running and never exactly aligned in centre frequency. The resulting beat appears as a linear phase ramp on the received samples. It is simulated by multiplying the signal by $\exp(j\,2\pi\,\Delta f\,T\,k)$, where Δf is the frequency offset, T the sample period, and k the sample index.

4.2.2 Isolating CFO

All frequency recovery algorithms first require that the phase of the data be removed before estimation. For QPSK symbols $c[i] = e^{j\frac{\pi}{4}+m[i]\frac{\pi}{2}}$, this can be done by raising each received symbol $x[i] = c[i]e^{j(2\pi f_d i+\theta[i])} + n[i]$ to the fourth power to give

$$z[i] = x^4[i] = e^{j(8\pi f_d i+4\theta[i])} + n'[i] \tag{4.1}$$

where θ is phase noise, f_d is the CFO, and n is additive noise. This allows all data symbols to be used in CFO estimation, but amplifies phase and additive noise, which degrades the estimation quality. The fourth power operation also requires three complex multiplications. An alternative is to use pilot symbols where the data phase is known beforehand, giving

$$z[i] = x[i]c^*[i] = e^{j(2\pi f_d i+\theta[i])} + n[i], \tag{4.2}$$

which is more robust to noise and less computationally expensive. Some algorithms only require $\arg(z[i])$, which is efficiently found using CORDIC as

$$\arg(z[i]) = \{4\arg(x[i])\}|_{-\pi}^{\pi} \tag{4.3}$$

for blind operation or

$$\arg(z[i]) = \{\arg(x[i]) - \arg(c[i])\}|_{-\pi}^{\pi} \tag{4.4}$$

for data-aided operation, where the wrap operation can be done with no additional cost by aligning the wrap range to the overflow range of the fixed-point value used to store the phase.

4.2.3 Rife & Boorstyn

The value of the CFO will appear as the strongest peak in the frequency spectrum of $z[i], z[i+1], \ldots, z[i+L]$ where L is the observation length. The Rife & Boorstyn algorithm first performs an FFT on the observed sequence and finds the index k corresponding to the strongest peak. This coarse estimation is refined with a parabolic interpolation on the complex FFT values X_{k-1}, X_k, X_{k+1} to give [11]

$$\delta = -\Re\left[\frac{X_{k+1} - X_{k-1}}{2X_k - X_{k-1} - X_{k+1}}\right]. \tag{4.5}$$

The updated index $k' = k + \delta$ is then scaled by the sampling rate to find the true CFO. The resolution in k is $\epsilon(k) = 1/2N_{FFT}$. The CPON specification provides only $L = 11$ training symbols, so data-aided operation without zero-padding gives $\epsilon(\hat{f}_d) \approx 0.05\,R_s$ — too coarse for the downstream phase recovery to correct. Zero-padding the training sequence to a larger N_{FFT} improves the resolution without requiring additional training data. The normalised mean squared-error (MSE) in the estimated CFO is shown in fig. 4.1 for data-aided operation with 11 training symbols. A CFO estimate is computed for both polarisations and averaged to reduce the MSE. The modified Cramer-Rao bound (MCRB) provides a lower bound on the achievable MSE and is given by [4]

$$\mathrm{MCRB} = \frac{3}{2\pi^2 L^3 \times \mathrm{SNR}}. \tag{4.6}$$

Rife & Boorstyn achieves this bound above a threshold SNR of ≈ 4 dB, where the threshold SNR is defined as the SNR below which the estimator begins to deviate from the MCRB. Below this, the noise has a significant contribution to the FFT values and can lead to an incorrect index being chosen.

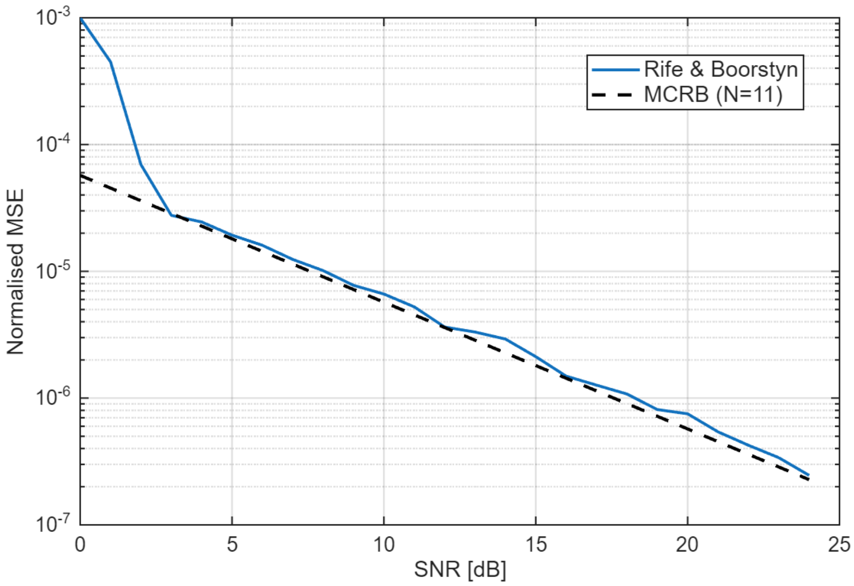


Figure 4.1: Normalised MSE of the Rife & Boorstyn algorithm with FFT size $N_{FFT} = 512$ compared to the MCRB for $L = 11$ training symbols.

Data-aided operation is constrained by the length of the training sequence, but the accuracy of the Rife & Boorstyn algorithm improves with longer observation lengths, as shown in fig. 4.2 for non-data-aided operation. The SNR threshold at which this algorithm begins to deviate from the MCRB is initially higher with data-aided operation due to the amplification of noise, but decreases for longer observation lengths.

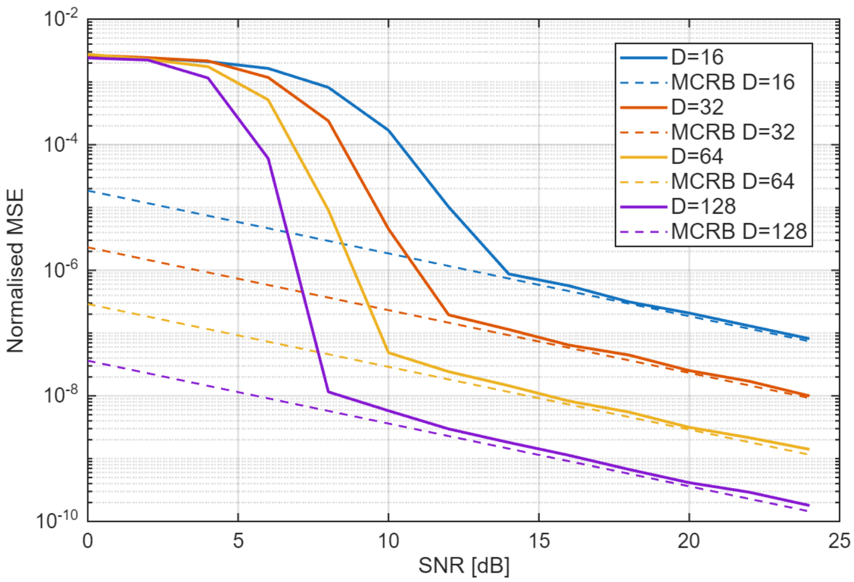


Figure 4.2: Normalised MSE of the Rife & Boorstyn algorithm for non-data-aided operation and different observation lengths.

4.2.4 Differential Phase and Kay Estimator

The Rife & Boorstyn algorithm exhibits good performance even at low SNRs, but is computationally expensive. The FFT requires $\frac{N_{FFT}}{2}\log_2(N_{FFT})$ complex multiplications and $N_{FFT} \log_2(N_{FFT})$ complex additions. The fine search from eq. (4.5) also requires a complex division. A cheaper approach would involve computing the phase difference

$$\arg(z[i]) - \arg(z[i-1]) = 2\pi f_d T_s + \theta[i] - \theta[i-1] + \eta[i] \tag{4.7}$$

over several symbols and averaging to remove the effect of θ. Estimating f_d in this manner is vulnerable to the effect of additive noise on the phase $\eta[i]$, so exhibits poor accuracy. This can be improved with a second pass using the Kay estimator. The Kay estimator uses a least-squares approach, computing the estimated CFO as

$$\hat{f}_d = \frac{R_s}{2\pi} \sum_{k=1}^{L} \frac{6k(L-k)}{L(L^2-1)} \arg(z[k]z^*[k-1]). \tag{4.8}$$

where $\arg(z[k]z^*[k-1])$ computes the phase difference from eq. (4.7) and $\frac{6k(L-k)}{L(L^2-1)}$ is a weighting term that prioritises the contribution of phase differences in the middle of the observed sequence. This estimator is able to achieve the MCRB, but the least-squares approach considers phase differences of non-adjacent symbols, making it vulnerable to phase wrapping that distorts the estimate when the CFO is high enough. When the total phase drift $\Delta\phi = 2\pi f_d L T_s$ across the observation window exceeds 2π, a wrap shifts the estimated CFO by $\pm 1/(LT_s)$. A first-pass coarse CFO correction must therefore be applied before the Kay estimator to keep the residual CFO within the unambiguous range. The performance of the differential phase and Kay estimator for data-aided operation is shown in fig. 4.3. The MCRB is achieved above a threshold SNR of ≈ 7 dB. This is higher than for Rife & Boorstyn, but at a reduced computational cost. Unlike the Rife & Boorstyn algorithm, the threshold SNR of the differential phase and Kay estimator does not depend on observation length, as shown in fig. 4.4.

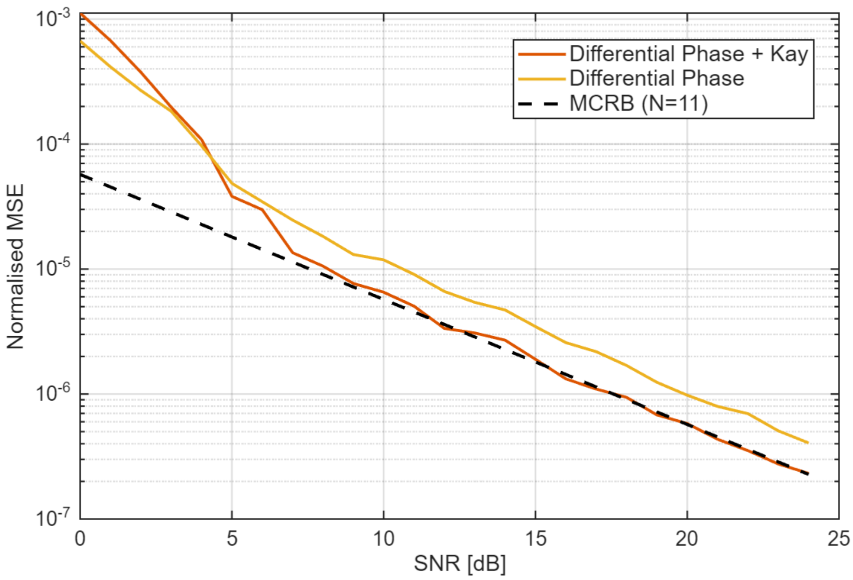


Figure 4.3: Normalised MSE of frequency estimation with the differential phase and Kay estimator for $L = 11$ training symbols.

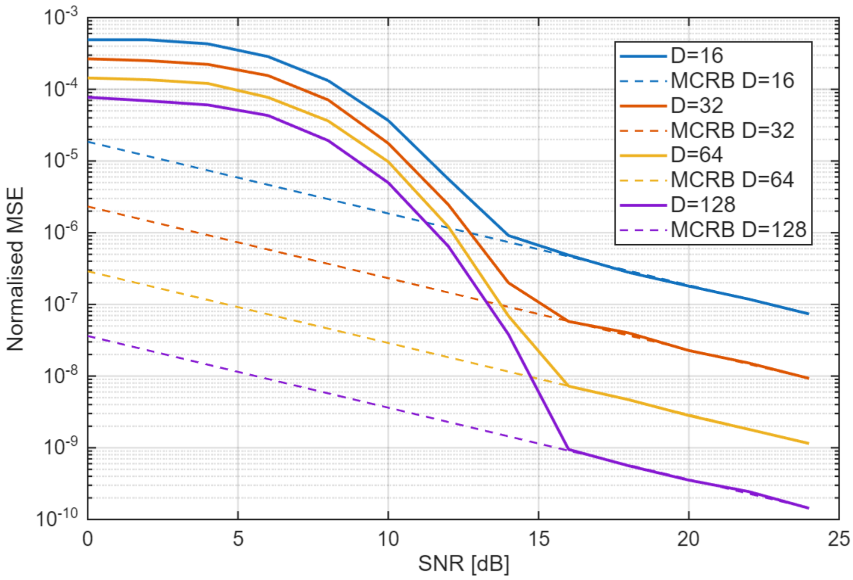


Figure 4.4: Normalised MSE of the differential phase and Kay estimator with blind operation for different observation lengths.

4.2.5 Computational Complexity

Here, the computational complexity of both frequency recovery algorithms is compared in terms of real multiplications and real additions. CORDIC is used to find phases and is assumed to cost the same as a real multiplication. The real division required by the R&B parabolic interpolation is assumed to cost the same as a multiplication since it is only performed once per estimate; a look-up-table reciprocal would meet this approximation at a small accuracy cost. All analysis applies to a single polarisation. If averaging is done over both polarisations, then all operations are doubled.

Rife & Boorstyn The Rife & Boorstyn algorithm requires an FFT followed by N_{FFT} comparisons of the magnitudes in the coarse search. Many of the FFT entries are 0 due to padding, which reduces the total number of operations needed. The interpolation requires the real part of a complex division

$$\Re\left[\frac{a+bj}{c+dj}\right] = \frac{ac+bd}{c^2+d^2}. \tag{4.9}$$

The total complexity is summarised in table 4.1; entries marked with a dagger (†) apply only to blind operation.

Operation	RM	RA
Forming $z[i], \ldots, z[i+L]$	$4L,\ 12L^\dagger$	$3L,\ 9L^\dagger$
FFT	$2L\log_2(\frac{N_{FFT}}{L}) + 2N_{FFT}\log_2(L)$	$3L\log_2(\frac{N_{FFT}}{L}) + 3N_{FFT}\log_2(L)$
Search	$2N_{FFT}$	$2N_{FFT}$
Interpolation	5	4

Table 4.1: Computational complexity of estimating CFO with the Rife & Boorstyn algorithm.

Differential phase and Kay The differential phase and Kay estimator has lower complexity, only requiring CORDIC, real additions, and real multiplications. The fixed multiplications by the weights of the Kay estimator can be optimised into a low-energy shift-add circuit, but this is ignored for simplicity. The first-pass CFO correction involves forming the phase corrections of each training symbol as $\phi[i] = \hat{f}_d\, i$, then applying them with CORDIC.

Operation	RM	RA
Forming $\arg(z[i]), \ldots, \arg(z[i+L])$	$2L$	L
Differential Phase	1	$L-1$
First Pass Correction	$2L$	0
Forming $\arg(z[i]), \ldots, \arg(z[i+L])$	$2L$	L
Kay	L	$L-1$

Table 4.2: Computational complexity of estimating CFO with the differential phase and Kay estimator.

4.2.6 Optimising Bit Width

These frequency recovery algorithms compute an estimate of the normalised CFO f_d/R_s. For n bits of precision, the smallest normalised CFO that can be estimated is 2^{-n} for an estimation range $0 \leq |f_d/R_s| \leq 1$. In reality, a much smaller range of CFOs needs to be estimated. If the maximum permitted value of $|f_d/R_s| = 2^{-d}$, d bits of precision that would otherwise be redundant can be utilised by scaling each estimate by 2^d during calculation, then inserting d bits in the final answer to recover the true normalised CFO. This improves the resolution to $2^{-(n+d)}$ without any increase in energy consumption.

4.2.7 Maximum CFO

The theoretical limit on the maximum CFO that can be corrected is given by Nyquist as $0.5\,R_s$. From eq. (4.1), we see that blind operation reduces this limit by a factor of 4. Both the Rife & Boorstyn and differential phase and Kay estimators accurately estimate the CFO up to the Nyquist limit, as seen in fig. 4.5. Blind operation aliases as expected at $f/R_s = 0.125$, making it viable for CFOs up to 3.8 GHz at a symbol rate of 30.5 GBd.

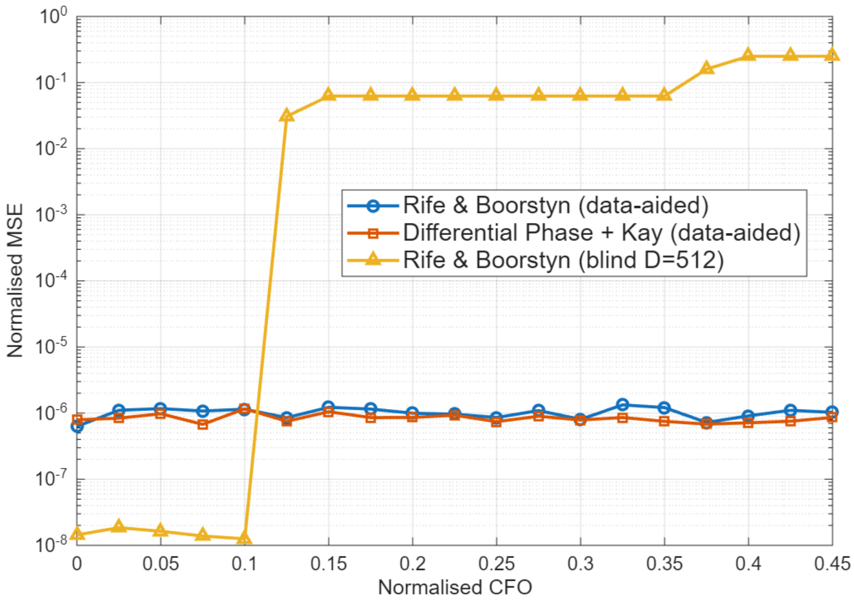


Figure 4.5: Normalised MSE in CFO estimate (MSE/R_s^2) vs. normalised CFO (f/R_s).

4.3 Phase Recovery

4.3.1 Phase Noise

Phase noise originates from the finite linewidth of the transmitter and LO lasers, whose phases follow a random walk. It is modelled as a Wiener process: independent Gaussian per-symbol phase increments with variance $\sigma^2 = 2\pi\,\Delta\nu/R_s$, where $\Delta\nu$ is the combined laser linewidth, are accumulated and applied as a multiplicative phase $\exp(j\phi[k])$.

4.3.2 Feed-Forward vs PLL Algorithms

One of the first techniques investigated for phase recovery was a digital phase-locked loop (PLL) to incrementally correct the phase of each symbol with feedback. This dramatically simplifies the phase estimation and correction due to the use of small-angle approximations. The CPON specification sets the symbol rate at 30.5 GBd. With an oversampling rate of 8/7, the total samples per second would be 35 GSa/s. This is beyond the achievable clock speed of CMOS, so symbols must be processed in blocks of e.g., 32 symbols with a clock speed of 1.1 GHz. This approach hinders the effectiveness of a PLL, as updates can only be made every 32 symbols. For this reason, feed-forward algorithms that can operate on blocks of data are used instead.

4.3.3 Viterbi & Viterbi

The Viterbi & Viterbi algorithm is well suited to QPSK and removes data phase with the fourth power operation of eq. (4.1). It then applies a maximum-likelihood filter $w_{ML}[i]$ to estimate the phase error due to Wiener phase noise from the combined laser linewidth and any residual CFO as

$$\hat{\theta}_{ML}[i] = \frac{1}{4}\arg\left(\sum_{k=1}^{N} w^*_{ML}[k]z[k]\right) - \frac{\pi}{4}. \quad (4.10)$$

A phase unwrapper is then applied before the phase error is corrected. This is readily adapted to block-based processing. If blocks of 32 symbols are used again, the ML filter can be applied to all symbols in the block and the estimated phase error applied to the entire block. This is only valid if the phase difference between the start and end of the block is small, meaning the laser linewidth cannot be too large and the preceding frequency recovery algorithm must correct most of the CFO. By increasing the time between phase estimates, block processing also increases the chance of cycle-slips in the phase unwrapper. The CPON specification provides one pilot symbol in every 32-symbol block that can be used to correct this [3]. The correction estimates the number of $\pi/2$ phase jumps n that occur between the unwrapped phase of a data symbol $\hat{\theta}_{\mathrm{PU}}$ (the output of the phase unwrapper) and its corresponding pilot, then adjusts the unwrapped estimate as $\hat{\theta}'_{\mathrm{PU}} = \hat{\theta}_{\mathrm{PU}} - n\frac{\pi}{2}$.

4.3.4 Pilots-Only Correction

While Viterbi & Viterbi is commonly used for QPSK, a simpler approach is to compute the phase error from the pilot symbol using eq. (4.2) and apply this to the whole block. This also relies on a small phase change across the block and is more vulnerable to additive noise. However, QPSK allows for $\pm\pi/4$ error in the phase estimate before an incorrect decision is made, which makes it robust to the drop in accuracy associated with this approach.

4.3.5 Computational Complexity

Viterbi & Viterbi For a block-based approach, the length of the filter will be the block length. Phases are again found with CORDIC. The cost of the phase estimate for each block is shown in table 4.3.

Operation	RM	RA
Forming $z[i], \ldots, z[i+N]$	$12N$	$9N$
Computing $\hat{\theta}_{ML}$	$4N+1$	$3N+2(N-1)+1$

Table 4.3: Computational complexity of block-based Viterbi & Viterbi.

Pilots-Only Only phases are needed, so the entire estimation can be performed with CORDIC. The pilot phase is also known beforehand, so does not contribute to the computational cost. The total cost of the phase estimate is shown in table 4.4.

Operation	RM	RA
Forming $\arg(z[i]) = \hat{\theta}$	1	1

Table 4.4: Computational complexity of Pilots-Only correction.

4.4 Methodology

4.4.1 Phase Recovery Comparison

The two phase recovery algorithms are compared in isolation by simulating an AWGN channel with additive Wiener phase noise, without chromatic dispersion, PMD, or fibre nonlinearity, so that any performance difference is attributable solely to the phase estimation stage. BER is estimated by Monte Carlo simulation over independent realisations for each combination of laser linewidth and SNR, with the phase ambiguity inherent to QPSK resolved by exhaustive search over the four $\pi/2$ rotations before symbol decisions are made.

4.4.2 System Energy Comparison

The full carrier recovery chain is then characterised over the same isolated channel, now with a fixed 3 GHz carrier frequency offset and Wiener phase noise at the 1 MHz linewidth limit of section 1.3.3, so the result depends only on frequency and phase recovery. Each frequency recovery algorithm is paired with each phase recovery algorithm, and for every pair the pre-FEC BER is estimated by Monte Carlo simulation over 100 independent subframes at each SNR from 0 to 30 dB. The SNR at the FEC limit is found by interpolation between adjacent SNR samples. It is computed per trial and reported as a mean with standard deviation.

Receiver energy per bit is evaluated from the per-symbol operation counts of tables 4.1 to 4.4 using the arithmetic energy model of section 2.2.1, with the frequency-recovery cost amortised over one 3712-symbol subframe and the phase-recovery cost over one 32-symbol block.

4.5 Results & Discussion

4.5.1 Phase Recovery Performance

To replace Viterbi & Viterbi with the simplified pilots-only phase correction, it must exhibit comparable performance over the range of linewidths that could be expected during operation of the network. The BER for each phase recovery algorithm as SNR varies is shown in fig. 4.6. Both algorithms have almost identical performance (plots overlapping) and are reliable up to 10 MHz, well beyond the maximum of 1 MHz given in the CPON spec.

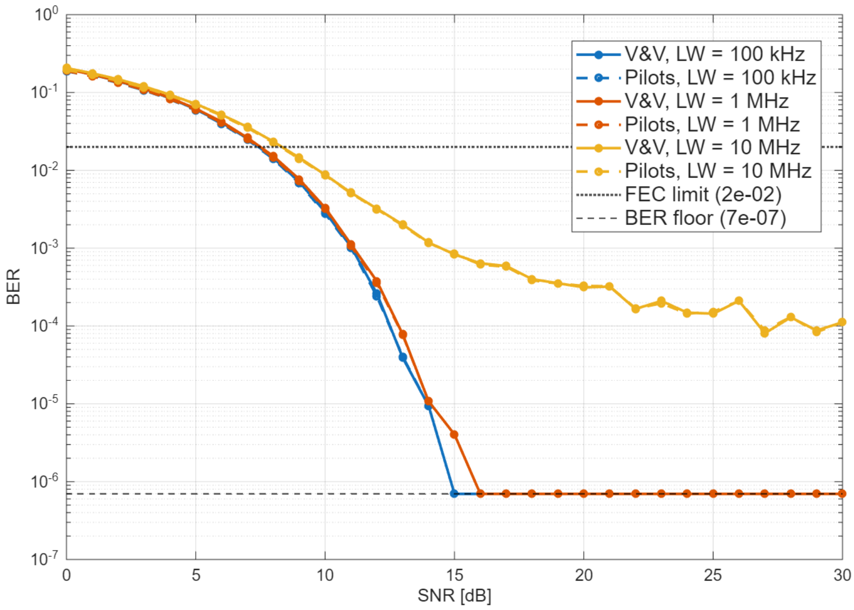


Figure 4.6: Performance comparison of V&V and Pilots-Only phase correction

The choice of fixed-point precision used in phase recovery will balance receiver energy and performance. Figure 4.7 compares the performance of both algorithms which is almost identical down to 6 bits, where the SNR at the FEC limit of Viterbi & Viterbi rises quickly. The additional operations needed by Viterbi & Viterbi to estimate the phase error mean that quantisation errors compound, degrading performance at low precision. The ability of the pilots-only estimate to match or improve on the SNR at the FEC limit of Viterbi & Viterbi is likely due to the inherent robustness of QPSK to additive and phase noise; pilots-only will produce a slightly less accurate estimate but as long as it brings the symbol within the correct QPSK quadrant, decoding will be successful.

It is then clear from fig. 4.7 that a receiver can save energy at no performance cost by using the pilots-only estimate with 4 bits of precision. The increase in transmitter energy from the slight increase in the SNR at the FEC limit at 2 bits of precision may be outweighed by the receiver energy savings, but this will depend on whether the carrier recovery block limits the receiver SNR and other system parameters like the wall-plug efficiency of the laser. All further investigations in this section use pilots-only for phase recovery.

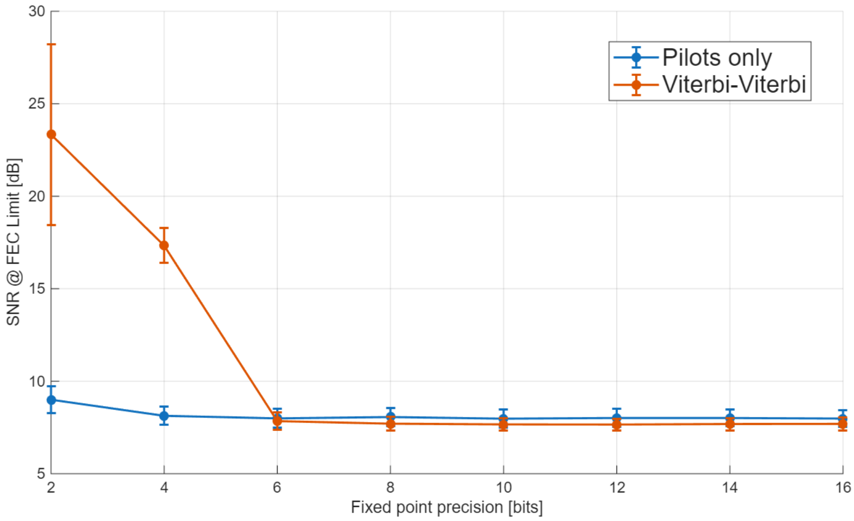


Figure 4.7: Phase recovery performance vs fixed-point precision at 1 MHz laser linewidth, with Rife & Boorstyn used to remove the initial CFO.

4.5.2 Blind vs Data-Aided Operation

The decrease in threshold SNR for longer observation lengths suggests that the performance of the Rife & Boorstyn algorithm will improve with blind observation length D. This is not the case for the differential phase and Kay estimator, where increasing blind observation length has no effect on threshold SNR. The effect of observation length on the SNR at the FEC limit is shown for both algorithms in fig. 4.8, where blind Rife & Boorstyn shows a clear increase in performance over the blind differential phase and Kay estimator for longer observations and improves on data-aided Rife & Boorstyn after ≈ 50 symbols. The differential phase and Kay estimator in blind operation is thus not viable for frequency recovery, but blind Rife & Boorstyn can provide the best performance at a higher computational cost. Here it is worth noting the impact of the higher energy of pilot and training symbols; by being placed at $\pm 3 \pm 3j$ they have $9/2 = 4.5\times$ greater energy than data symbols. These high energy symbols increase the average energy of a transmission by ≈ 0.4 dB but allow data aided estimators to attain the same accuracy at an SNR of ≈ 6.5 dB lower than if all symbols were placed at $\pm 1 \pm j$.

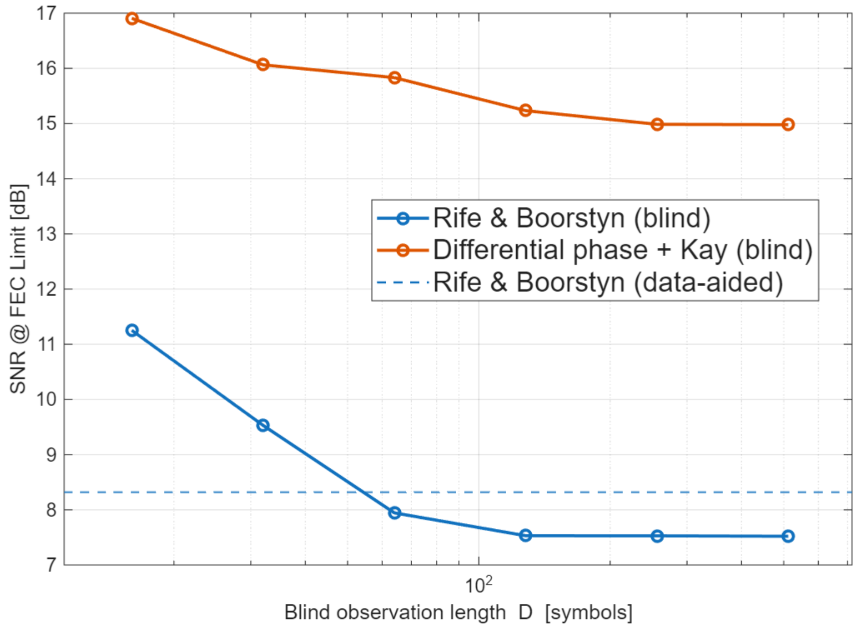
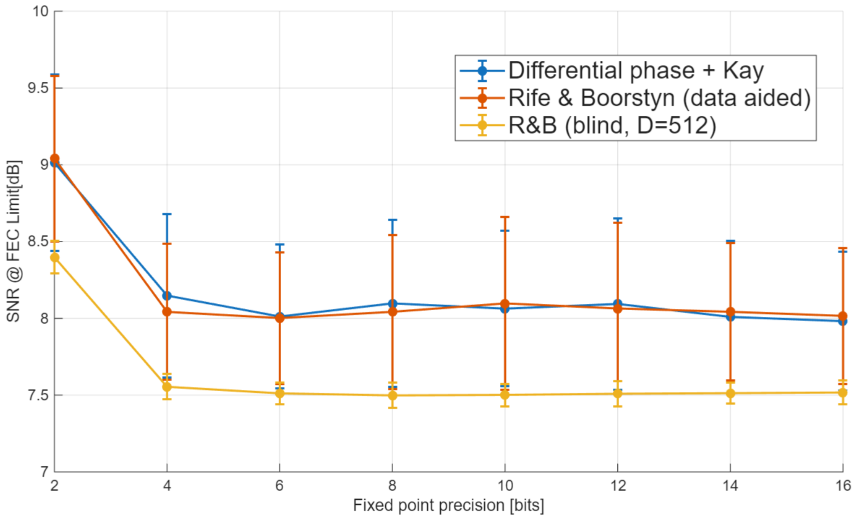


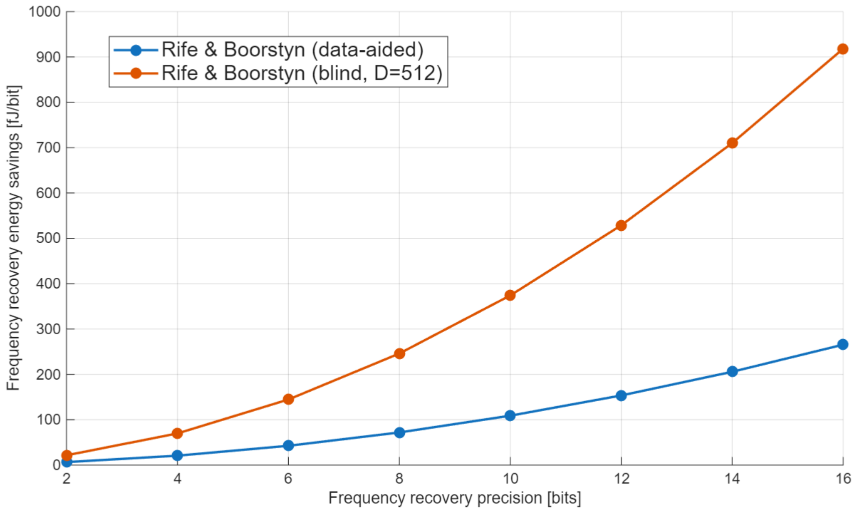
Figure 4.8: SNR at the FEC limit vs blind observation length for the Rife & Boorstyn and differential phase and Kay estimators.

4.5.3 Frequency Recovery Performance

The fixed-point performance of each frequency recovery algorithm is compared in fig. 4.9a, and the corresponding energy savings when switching to the differential phase and Kay estimator are shown in fig. 4.9b. Data-aided Rife & Boorstyn appears to offer no performance improvement on differential phase and Kay but consumes more energy. Blind Rife & Boorstyn consumes the most energy but offers a significant performance increase over both data-aided algorithms. The performance of all algorithms does not degrade until 4 bits of precision. Decreasing the precision further will again depend on the balance between transmitter and receiver energy. The most energy-efficient algorithm depends on how frequently the CFO must be estimated; frequent estimates will likely favour the differential phase and Kay estimator but in the extreme case of only one CFO estimate per transmission then amortisation makes the energy consumption negligible and blind Rife & Boorstyn will be preferred. Only using 11 symbols in the CFO estimate for data-aided operation introduces high sensitivity to the noise present in those symbols, hence a high variance in the SNR at the FEC limit is observed. This has significant design implications as a transmitter tailored to provide the average SNR at the FEC limit at the receiver could cause decoding errors if the noise on the 11 training symbols is particularly high. A safe design would then ensure the receiver SNR is closer to the maximum measured SNR at the FEC limit. The variance could be reduced by averaging the CFO estimate over several subframes, but this requires the CFO to stay constant for long enough and consumes more energy at the receiver.



(a) SNR at the FEC limit.



(b) Energy saved by switching to the differential phase and Kay estimator

Figure 4.9: Frequency recovery as fixed-point precision varies, comparing data-aided differential phase and Kay, data-aided Rife & Boorstyn, and blind Rife & Boorstyn, with the corresponding energy savings when switching to differential phase and Kay. FR energy amortised over one subframe (3712 symbols).

4.5.4 Energy Breakdown

Blind Rife & Boorstyn and differential phase and Kay with pilots-only phase recovery are the two most promising approaches for energy-efficient carrier recovery. The energy breakdowns for each are shown in fig. 4.10, and the combined energies and SNRs at the FEC limit are listed in table 4.5. Estimating the phase error with pilots-only consumes very little energy, and blind R&B dominates energy consumption in estimating CFO even with amortisation over a subframe. Differential phase and Kay with 4 bits of precision offers a lower SNR at the FEC limit for less energy than blind Rife & Boorstyn with 2 bits of precision, so the most energy-efficient implementation will select one of the other 3 combinations depending on the balance between transmitter and receiver energy.

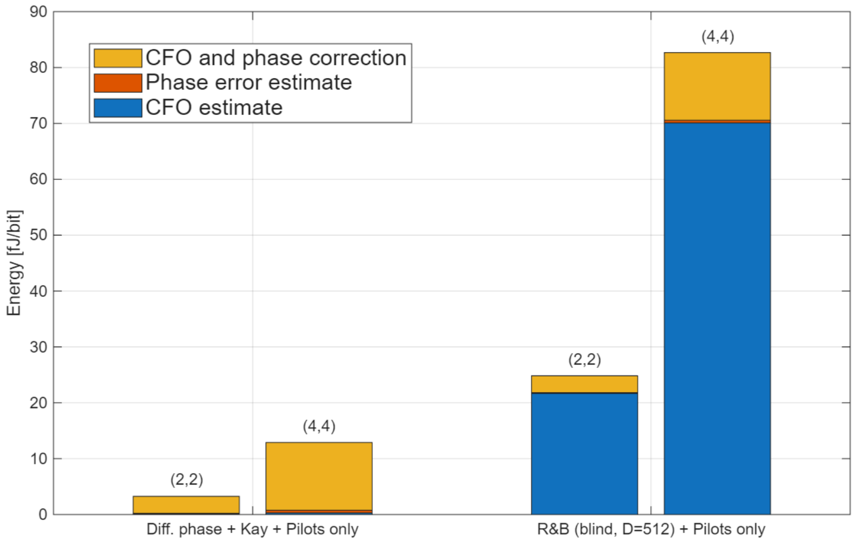


Figure 4.10: Receiver energy breakdown of carrier recovery for 4 bits (left) and 2 bits (right) of precision.

FR algorithm	Fractional bits	E_{FR} (fJ/bit)	E_{CR} (fJ/bit)	E_{apply} (fJ/bit)	E_{total} (fJ/bit)	SNR @ FEC Limit (dB)
Diff. phase + Kay	2	0.08	0.14	3.03	3.26	9.76 ± 0.89
	4	0.29	0.48	12.12	12.89	8.11 ± 0.50
R&B (blind, $D = 512$)	2	21.66	0.14	3.03	24.83	8.96 ± 0.16
	4	70.08	0.48	12.12	82.68	7.60 ± 0.09

Table 4.5: Combined energy and SNR at the FEC limit of the differential phase and Kay estimator and blind Rife & Boorstyn, both using pilots-only phase recovery.

Chapter 5

Full DSP Pipeline

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